Monitoring and Understanding Notebook-Based Learning with Jupyter

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Open-Minded

1. Motivation

Computational notebooks are widely used in programming and data science education. However:

- Final notebook submissions show the result, but not the learning process
- Students rarely receive feedback about their own debugging and execution behavior
- Existing learning dashboards often rely on coarse-grained LMS data rather than cell-level programming activity

2. Related Work

- Learning Analytics Dashboards often rely on coarse-grained LMS data and provide limited insight into the programming process [8, 9].
- Notebook Analytics: Existing approaches analyze fine-grained notebook activity, but are often instructor-facing, assignment-level, or retrospective [1, 6, 13].
- AI-supported Tools: Recent systems combine notebook analytics with AI-based tutoring and feedback [17].

→ **Our Focus**

- A learner-facing, dashboard that provides live, cell-level insights directly within JupyterLab to support student reflection.

3. System Design

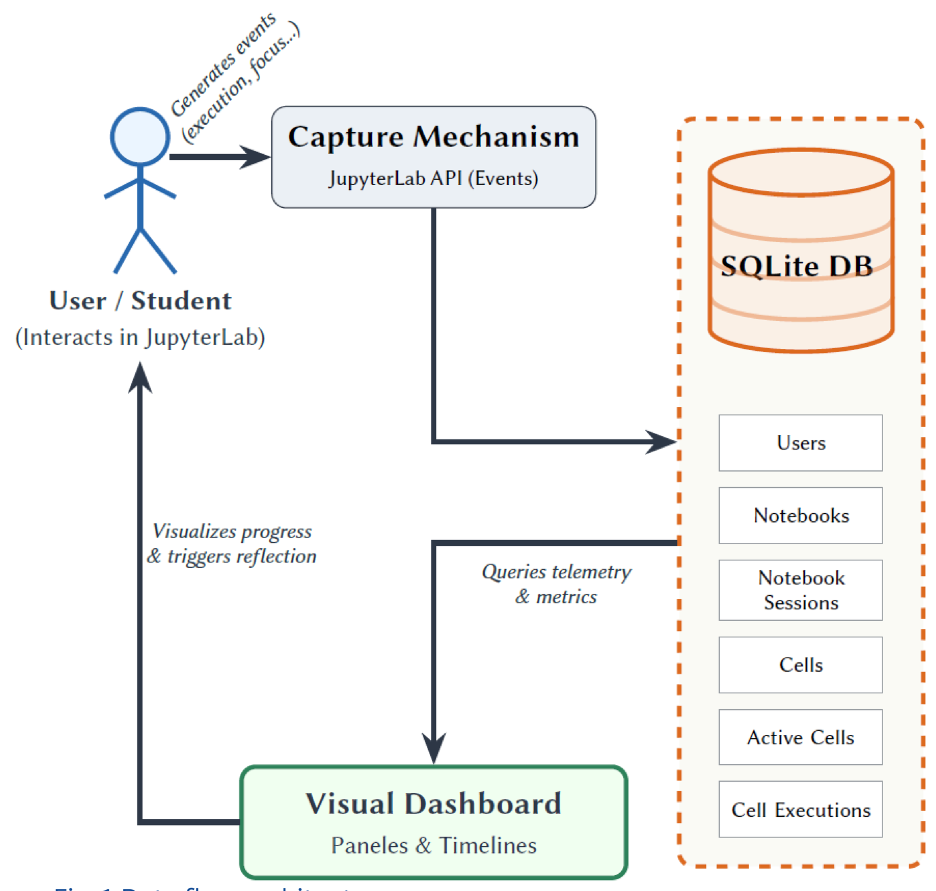


Fig. 1 Data flow architecture.

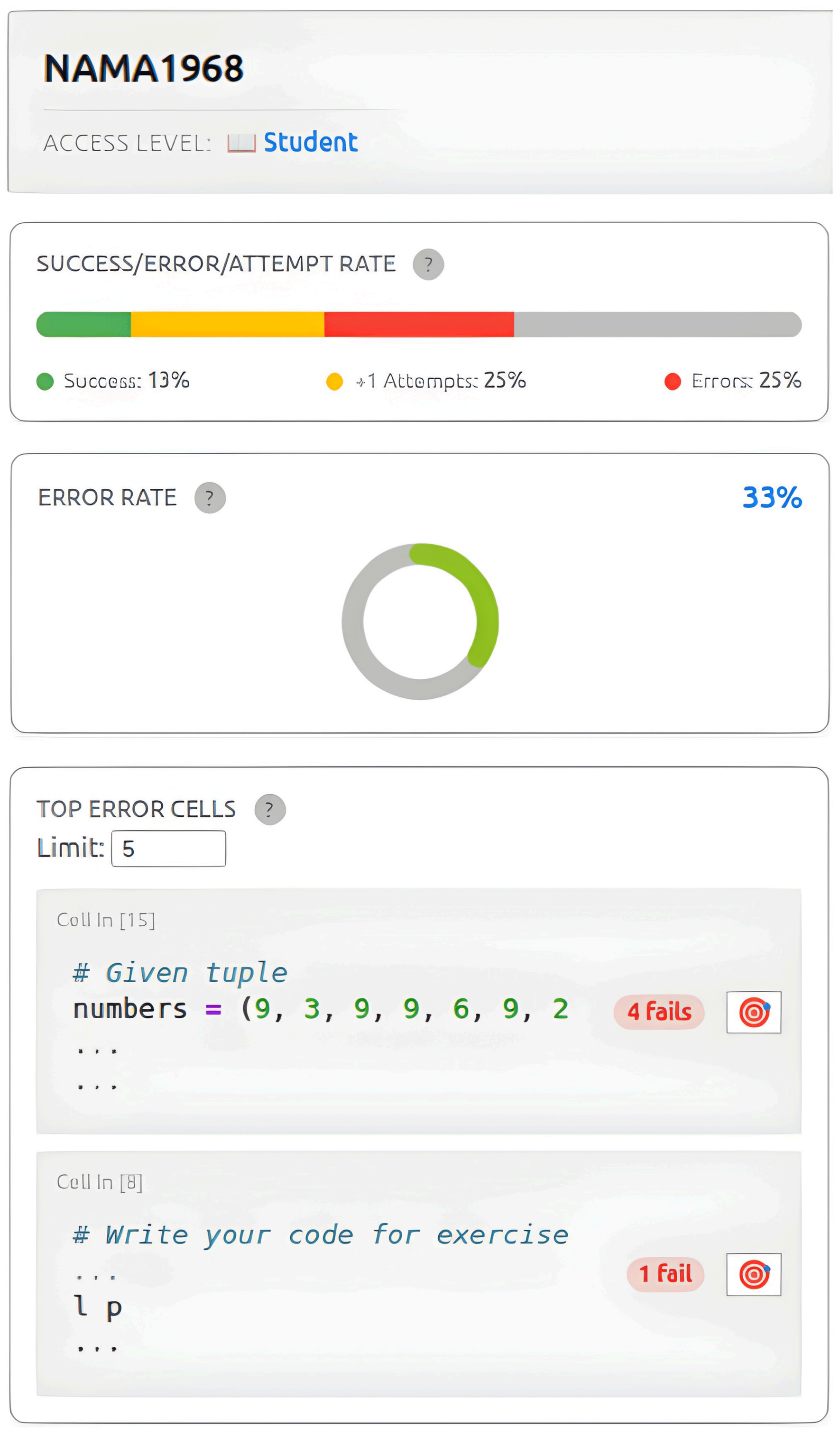


Figure 2: The upper section of the learner-facing JupyterLab dashboard.

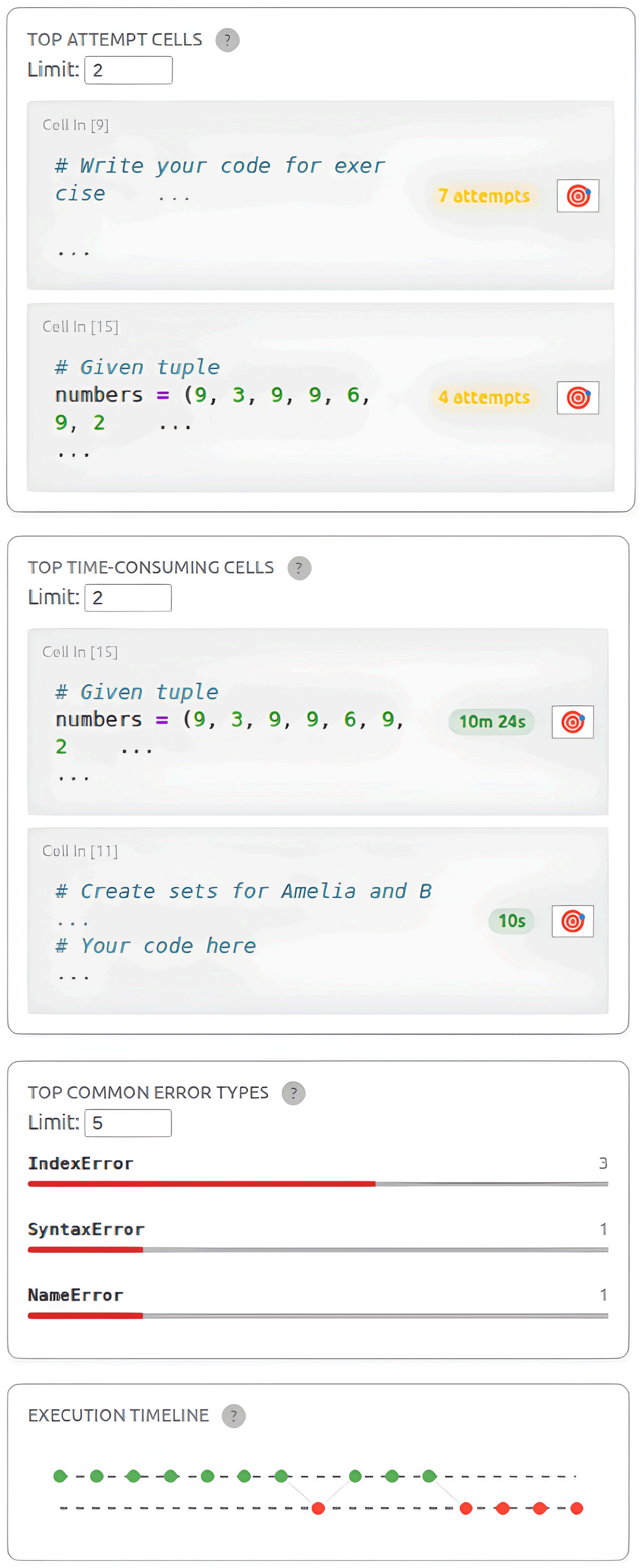
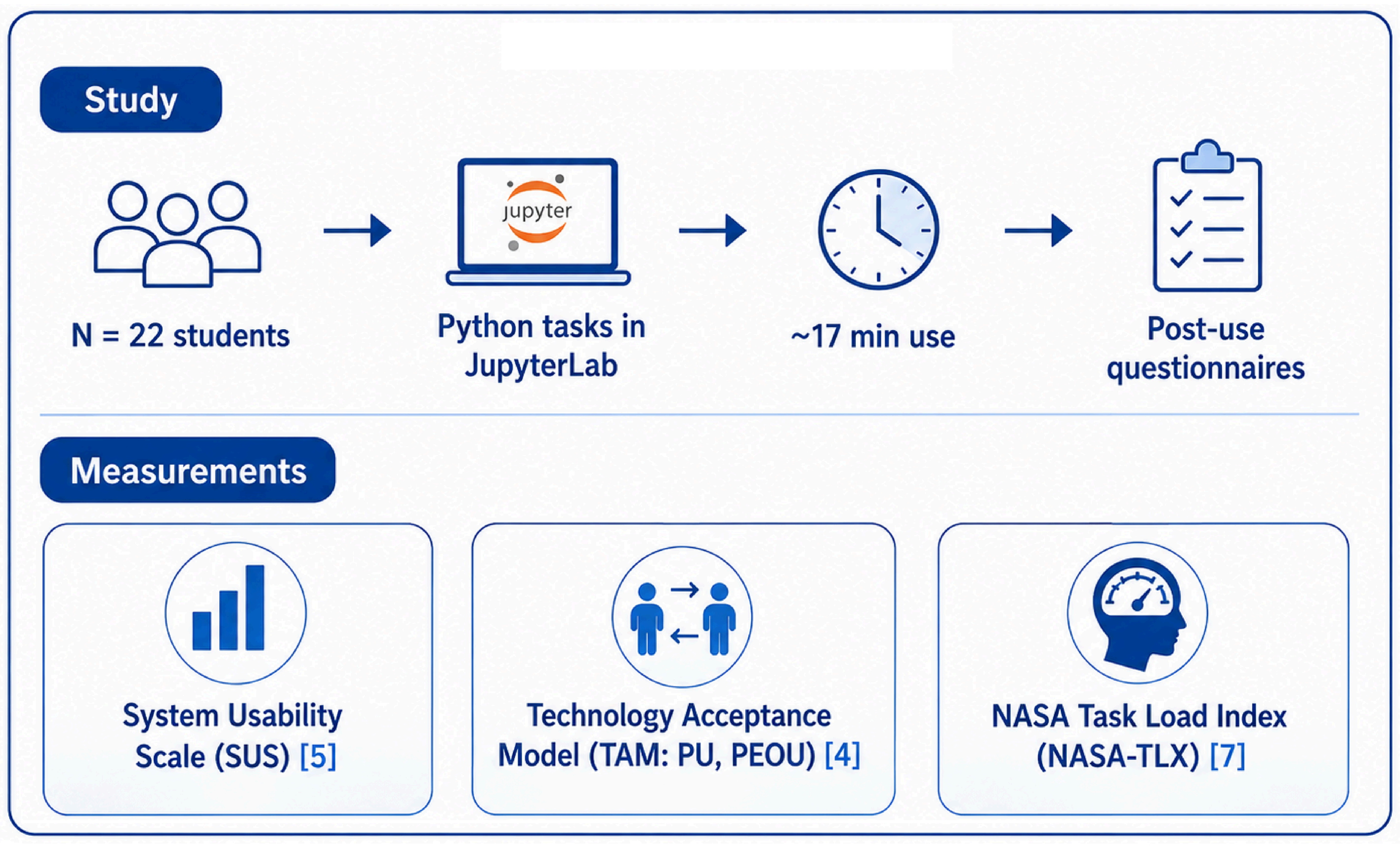


Figure 3: The lower section of the learner-facing JupyterLab dashboard.

4. Evaluation



5. Results

Table 1: Descriptive Statistics of Main Evaluation Measures (N = 22)

Measure	M	SD	Min	Max
System Usability Scale (SUS)	65.57	17.89	40.00	90.00
TAM Overall	5.36	0.88	3.25	6.75
Perceived Usefulness (PU)	4.93	1.18	3.33	7.00
Perceived Ease of Use (PEOU)	5.78	1.06	3.17	7.00
NASA-TLX	8.15	2.92	1.67	12.67

6. Discussion

Easy to integrate

→ Students found the dashboard relatively easy to use, with low-to-moderate workload.

Visibility alone is insufficient

→ Visualizing errors and execution patterns does not necessarily support meaningful reflection.

Actionable support is needed

→ Explanations and concrete recommendations may help students translate dashboard insights into next steps.

7. Conclusion

Fine-grained notebook analytics can make students' programming processes visible - but visibility alone is not enough.

Learners need actionable guidance to translate activity patterns into meaningful next steps.

Future work: Longitudinal and controlled studies should examine effects on debugging, self-reflection, programming strategies, and learning outcomes.

References
[1] Cai et al., 2025 → Notebook Analytics / student activity in Jupyter
[4] Davis et al., 1989. Technology Acceptance Model (TAM)
[5] German UPA, 2026. System Usability Scale (SUS)
[6] Groher et al., 2024 → Learning Analytics Dashboard in programming classes
[7] Hart & Staveland, 1988. NASA Task Load Index (NASA-TLX)
[8] Jivet et al., 2020 → learner-facing dashboards, sense-making / reflection
[9] Paulsen & Lindsay, 2024 → systematischer Review zu Learning Analytics Dashboards
[13] Raghunandan et al., 2023 → retrospective analysis of notebook changes
[17] Valle Torre et al., 2025 → AI + Learning Analytics in Jupyter

Video of the extension

https://colaps.team/